

Class-Aware Data Augmentation for Long-Tailed Object Detection on the BadODD Dataset

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Abstract—Long-tailed class distributions are a persistent obstacle to reliable object detection in autonomous driving, because aggregate performance metrics are dominated by frequent categories while rare but safety-relevant categories may fail entirely. This paper examines that problem on BadODD, a Bangladeshi road-scene detection dataset whose training split exhibits an imbalance ratio of 869.9 between its most and least frequent categories, with the three most frequent accounting for over seventy percent of all annotated instances. Two data-centric interventions are evaluated on a fixed YOLO11s detector: class-aware oversampling of training images containing under-represented categories, and targeted bounding-box-safe augmentation applied only to those images. Both configurations were trained under an identical protocol on a single Tesla T4 with a fixed seed and a 25-epoch budget, and evaluated once on a held-out test split of 2,033 images. The combined strategy raised mean average precision at IoU 0.50 from 60.78 to 62.86 and at IoU 0.50:0.95 from 38.58 to 40.24, while precision fell from 75.41 to 73.30 and recall from 54.88 to 53.82. The gain was concentrated in the targeted group, whose mean average precision at IoU 0.50 rose by 6.83 points against 0.03 points for the most frequent categories. Comparable changes in untargeted categories, combined with a single training seed, mean these results are consistent with the hypothesis but do not establish it.

Index Terms—object detection, long-tailed distribution, data augmentation, class-aware sampling, autonomous driving, BadODD

I. INTRODUCTION

Object detection is a foundational component of autonomous driving perception. A driving system must localise and categorise the objects around it before it can plan a trajectory, and errors of omission at this stage propagate through every downstream module. The reliability of detection across the full range of objects a vehicle may encounter is therefore a safety property, not merely a performance metric.

The visual composition of road scenes in Bangladesh differs substantially from that of the environments in which most autonomous driving benchmarks were collected. Traffic is heterogeneous and densely mixed, and the vehicle population

includes categories that are rare or absent in Western and East Asian datasets, such as auto-rickshaws, three-wheelers and cart-vehicles. Detectors trained elsewhere cannot be assumed to transfer to this setting.

BadODD was assembled to address this gap [1]. It comprises road scenes collected across nine districts of Bangladesh under both day and night conditions and spanning urban, rural, highway and expressway environments, annotated across thirteen object categories [1], and is distributed publicly [2]. Its coverage makes it a suitable basis for studying detection under the conditions that a deployed system in this region would actually encounter.

Like most naturally collected detection datasets, BadODD is severely long-tailed. In the training split examined in this work, the three most frequent categories account for over seventy percent of all annotated instances, while the three least frequent account for less than one fifth of one percent, and the ratio between the most and least frequent class counts exceeds 850. This structure creates a specific evaluation hazard: aggregate mean average precision is dominated by the categories that supply most of the instances, so a detector can report a respectable aggregate score while failing outright on the categories that occur rarely. Several of the rare categories in BadODD — construction vehicles, wheelchairs — are precisely those whose detection matters most for vulnerable road user safety.

Existing work on BadODD has concentrated on establishing the dataset and on comparing detector architectures. The original release characterised the data and reported baselines from the YOLO family [1], and subsequent studies have applied transformer-based detectors to Bangladeshi road scenes [3], [4]. This body of work establishes what different architectures achieve on the dataset. It does not isolate the effect of data-centric training interventions on the dataset's specific class distribution, and the question of whether simple changes to how training images are sampled and augmented can improve

rare-category detection on BadODD remains open.

This paper addresses that question through a controlled empirical study. Two data-centric interventions are applied to a fixed detector: class-aware oversampling of training images that contain under-represented categories, and targeted bounding-box-safe augmentation applied only to those same images. Neither intervention modifies the architecture, the loss function or the optimisation procedure, and both are reproducible from the dataset statistics alone. The study is deliberately narrow in scope: it is a dataset-specific empirical evaluation of simple, reproducible class-aware training strategies on BadODD, not a proposal for a general long-tailed detection algorithm.

The contributions are as follows. First, a quantified characterisation of the long-tail structure of the BadODD training distribution using both instance frequency and image frequency, together with a data-driven rule for grouping categories by support, and documenting a discrepancy between the class counts reported in the original description of the dataset and those enumerated in the publicly distributed release. Second, a reproducible YOLO11s baseline under a fully documented and fixed training and evaluation protocol. Third, a controlled comparison between that baseline and the combined class-aware strategy in which every variable other than the training manifest is held constant; the compute budget did not permit the additional runs required to isolate the two components separately, and no claim is made as to which component is responsible for any observed difference. Fourth, a support-aware analysis in which every reported average precision is accompanied by its training and test instance counts, and categories with small test support are identified explicitly rather than averaged into a single rare-category figure.

The remainder of this paper is organised as follows. Section II reviews related work. Section III analyses the dataset and quantifies its imbalance. Section IV describes the proposed method. Section V specifies the experimental setup. Section VI reports results. Section VII discusses them, Section VIII states the limitations of the study, and Section IX concludes.

II. RELATED WORK

A. Object Detection Architectures

Modern object detection divides broadly into three families. Two-stage detectors generate region proposals before classifying and refining them, an approach established by the Faster R-CNN line of work [5]. Single-stage detectors dispense with the proposal stage and predict categories and box coordinates directly from a dense feature map, a formulation introduced by YOLO [6] and refined across successive generations that progressively improved the trade-off between accuracy and inference cost [7]. A third family reformulates detection as direct set prediction using transformers, removing hand-designed components such as anchor generation and non-maximum suppression [8], with subsequent work bringing this formulation into the real-time regime [9].

For a study whose object is the training data rather than the model, the detector functions as an instrument and should be

stable, well documented and inexpensive to train repeatedly under identical conditions. A small single-stage detector satisfies these requirements, and the present work uses YOLO11 [10] for this reason rather than on grounds of architectural superiority.

B. Road Object Detection in Bangladesh

BadODD was introduced to provide detection data representative of Bangladeshi road conditions, covering multiple districts, both daylight and night-time acquisition, and a category set that includes vehicle types specific to the region [1]. The original work characterised the dataset and reported baseline results using detectors from the YOLO family. Subsequent studies have applied transformer-based detectors to this setting, examining real-time detection transformer variants on Bangladeshi road data [3] and end-to-end transformer detection for vehicle recognition under difficult road conditions [4].

These studies share a common orientation: they establish the data and compare what different architectures achieve on it, reporting aggregate detection metrics. What they do not examine is whether the training procedure itself, holding the architecture fixed, can be adapted to the dataset's particular class distribution.

C. Long-Tailed Object Detection

Skewed category distributions have been recognised as a difficulty for detection for a considerable time, including in the context of fine-tuning pretrained models on data whose category frequencies differ sharply from those of the pretraining corpus [11]. The introduction of benchmarks with explicitly long-tailed category vocabularies formalised the problem and established evaluation protocols that report performance separately for rare categories rather than only in aggregate [12], and the field has since been surveyed comprehensively [13].

One major line of response modifies the loss function so that gradient contributions are rebalanced across categories. Focal loss reweights by example difficulty to address foreground-background imbalance [14]; equalisation loss addresses the suppression of rare-category gradients by frequent ones [15]; and seesaw loss adjusts the balance dynamically according to accumulated category frequency [16]. These methods are effective but require modification of the training objective, which places them outside the scope of a study designed to hold the model and its objective fixed.

D. Sampling Strategies for Rare Categories

The complementary line of response leaves the objective unchanged and alters what the model sees. Repeat-factor sampling increases the frequency with which images containing rare categories are presented during training, deriving the repetition rate from category frequency with dampening to prevent extreme oversampling [12], and later work refines this by accounting for the number of instances present rather than mere presence [17]. This distinction between image frequency and instance frequency is significant in practice, since a category may be rare in the number of images in which

it appears while occurring many times within each of them, or the reverse; the present work reports both quantities for this reason.

E. Data Augmentation

Data augmentation is a long-standing means of improving generalisation from limited data, and the range of available transformations and their effects have been surveyed extensively [18], with efficient implementations available in established libraries [19]. Modern detection pipelines incorporate augmentation directly into training, with mosaic composition among the techniques that became standard in the YOLO family [7]. Copy-paste augmentation, which composites object instances into new scenes, has been shown to be a strong technique in its original setting [20]; that setting was instance segmentation, where pixel-accurate masks make the composited boundaries clean, and its effectiveness there is not direct evidence for bounding-box detection.

F. Summary and Gap

Existing BadODD studies establish the dataset and compare detector architectures on it [1], [3], [4]. The long-tailed detection literature offers both loss-reweighting [14]–[16] and data-centric [12], [17] responses to skewed distributions, but has developed and evaluated them principally on large general-purpose benchmarks. The effect of image-level class-aware sampling and targeted augmentation on the specific distribution of BadODD, with the detector and every other training variable held fixed, has not been reported. This work addresses that gap.

III. DATASET AND PROBLEM ANALYSIS

A. Source and Composition

The BadODD dataset was introduced as a road-object detection benchmark for Bangladeshi driving scenes, collected across nine districts under day and night conditions and spanning urban, rural, highway and expressway environments [1]. This study uses the release deposited on Zenodo under DOI 10.5281/zenodo.13823687 [2]. The annotations are distributed in YOLO format, with one text file of normalised bounding boxes per image, covering thirteen object classes: auto-rickshaw, bicycle, bus, car, cart-vehicle, construction-vehicle, motorbike, person, priority-vehicle, three-wheeler, train, truck and wheelchair.

The copy obtained from the Zenodo deposit did not include a class-name mapping file. The class names were therefore recovered from the published description of the dataset [3] and the assignment of names to label indices was verified against the class-frequency table reported in [1], using the rank order and magnitude of the three rarest classes as the identifying criterion. This verification is a prerequisite for the per-class results in Section VI, since an incorrect index assignment would attach correct measurements to incorrect class labels.

TABLE I
BADODD STATISTICS: REPORTED VERSUS OBSERVED

Property	Reported [1]	Observed
Total images	9,825	10,032
Total annotated objects	78,943	80,036
Classes	13	13
Training images	5,896	5,967
Validation images	1,964	2,032
Test images	1,965	2,033
Train inst., construction-vehicle	23	24
Train inst., wheelchair	2	29
Train inst., train	1	21
Test inst., train	0	16

B. Reported and Observed Statistics

Table I compares the dataset statistics reported in the original description [1] with those observed by direct enumeration of the obtained copy. The observed totals exceed the reported ones by 207 images and 1,093 annotated objects, and the observed split sizes differ correspondingly. The discrepancy is not attributable to a different partitioning of the same data, since a repartition would redistribute instances between splits without altering class totals. Two classes account for most of the difference: the training split of the obtained copy contains 21 train and 29 wheelchair instances, whereas [1] reports 1 and 2 respectively, while construction-vehicle is closely consistent at 24 against a reported 23. Both sets of values are reported here; the Zenodo deposit postdates the original description, and no claim is made as to which values describe the intended release.

C. Long-Tail Characterisation

The training split contains 47,514 annotated objects across thirteen classes. The distribution is severely skewed. The three most frequent classes — person, auto-rickshaw and three-wheeler — account for 72.91 percent of all training instances, while the three least frequent — construction-vehicle, train and wheelchair — account for 0.16 percent. The imbalance ratio, defined as the ratio of the largest to the smallest non-zero class count, is 869.9. The coefficient of variation across class counts is 1.408. The most frequent class holds 9.69 times the median count, and the least frequent holds 0.011 times the median.

Instance frequency and image frequency diverge substantially across the distribution, and both are reported here because the two quantities motivate different sampling strategies [12], [17]. The person class appears 18,268 times across 4,807 training images, an average of 3.80 instances per image containing the class. Each of the three rarest classes appears close to once per image containing it: 1.09 for construction-vehicle, 1.11 for train, and 1.00 for wheelchair. As a consequence, the imbalance measured at image level is markedly milder than at instance level: the ratio of the largest to the smallest image frequency is 253.0, against an instance-level ratio of 869.9. Fig. 1 shows both quantities on a logarithmic scale.

A further property of the distribution is that the image sets of the three rarest classes are disjoint. Their image frequencies

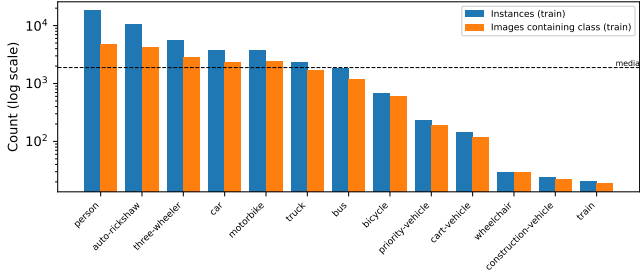


Fig. 1. BadODD training-split class distribution on a logarithmic scale, showing instance frequency and the number of images containing each class. The dashed line marks the median class count used in the repeat-factor computation.

of 22, 19 and 29 sum to 70, which equals the total number of training images containing at least one of them, so no training image contains more than one of these classes.

D. Data-Driven Class Grouping

Classes were grouped by quartiles of the positive training instance counts rather than by inspection. The quartiles are $Q_1 = 144$, median = 1,885 and $Q_3 = 3,786$. Classes with counts above Q_3 form the majority group, classes with counts in the closed interval $[Q_1, Q_3]$ form the medium group, and classes with counts below Q_1 and with non-zero test support form the minority group. A fourth group is reserved for classes whose support is too small for AP to carry information; the criteria adopted were a training count of at most 10 instances or a test count of at most 5 instances. These thresholds were fixed before any model was trained, on the basis of the training distribution alone, and no evaluation result was consulted in setting them.

The resulting assignment is: majority — person, auto-rickshaw, three-wheeler; medium — car, motorbike, truck, bus, bicycle, priority-vehicle, cart-vehicle; minority — wheelchair, construction-vehicle, train. Two classes fall exactly on a quartile boundary and are assigned to the medium group by the closed-interval rule: car at 3,786 instances equals Q_3 , and cart-vehicle at 144 equals Q_1 . No class in this copy of the dataset met the fourth-group criteria, and every class has non-zero test support, so all thirteen classes are evaluable.

E. Support Limitations

The three minority classes carry training supports of 29, 24 and 21 instances and test supports of 28, 11 and 16 instances. These counts are sufficient for AP to be computed, and none of the three requires exclusion on the grounds of zero support. They are nonetheless small enough that individual detection outcomes move the reported value substantially, and per-class results for this group are therefore accompanied by their support counts throughout Section VI and are not presented as established effects. This is a material difference from the distribution described in [1], where the train class has no test instances at all and would not be evaluable; in the obtained

copy every class can be evaluated, and the fourth group is empty.

IV. PROPOSED METHOD

A. Baseline Detector

All configurations use YOLO11s initialised from the COCO-pretrained checkpoint distributed by the model’s maintainers [10]. The choice of a small, well-documented single-stage detector is instrumental rather than architectural: the object of study is the training data, so the detector is held fixed across configurations and no modification is made to its architecture, loss function or optimisation procedure. No formal research publication describes YOLO11; the model is cited to its official documentation [10].

B. Component A: Class-Aware Image-Level Oversampling

Oversampling is applied at the level of whole images rather than by duplicating individual annotations. An object is embedded in a scene, and replicating an annotation without its surrounding context would produce training examples inconsistent with the images from which they derive.

The procedure enumerates the instance count and the image frequency of every class in the training split, groups classes by the rule of Section III, and forms the set of training images containing at least one minority-group class. A per-class repeat factor is computed with square-root dampening and a hard ceiling,

$$r_c = \min\left(r_{\max}, \sqrt{\frac{n_{\text{med}}}{n_c}}\right), \quad (1)$$

and the repeat factor of an image is the ceiling of the largest r_c among the minority-group classes it contains, with all other images assigned a factor of one:

$$r_{\text{img}} = \left\lceil \max_{c \in \mathcal{M} \cap \mathcal{C}(\text{img})} r_c \right\rceil. \quad (2)$$

The ceiling $r_{\max}$ was fixed at 4. The dampening term follows established repeat-factor practice in long-tailed detection [12], [17] and is an implementation choice, not a contribution of this work. Validation and test manifests are left unmodified.

Applied to this dataset, 70 of 5,967 training images (1.17 percent) contain a minority-group class and are selected for repetition. Because the three minority classes lie far below the median count, the uncapped repeat factors are 8.86, 9.47 and 8.06 for construction-vehicle, train and wheelchair respectively, all of which exceed the ceiling. The ceiling is therefore binding for every selected image, the square-root dampening has no effect on the resulting manifest, and every selected image receives a repeat factor of exactly 4. The manifest consists of 5,897 images at a factor of one and 70 images at a factor of four, yielding 210 additional image slots.

C. Component B: Targeted Bounding-Box-Safe Augmentation

Augmentation is applied offline and only to the selected minority-containing images, so that the majority of the training set is unaffected. Two variants are generated per selected image using the Albumentations library [19], with bounding

boxes transformed jointly with the image, re-clipped to the image bounds, and discarded when the transformed box falls below 30 percent of its original visible area or 16 pixels in area.

The transform set is deliberately conservative. Horizontal flipping is applied with probability 0.5, since road-object semantics remain valid under mirroring. Scale jitter of ± 20 percent and translation of ± 10 percent are each applied with probability 0.6, reflecting variation in apparent object distance and discouraging a fixed spatial prior. Rotation is restricted to ± 7 degrees with probability 0.5, since road geometry is predominantly upright. Brightness and contrast are varied by up to 20 percent with probability 0.5, and hue and saturation mildly with probability 0.25, reflecting the day and night conditions and the camera variation present in the dataset [1]. Gaussian blur and additive noise are applied at low strength with probability 0.1 each; stronger degradation was avoided because the rare objects of interest are often small and are destroyed by aggressive blurring.

Copy-paste augmentation was not used as a component of the method. The current documentation identifies it as a segmentation-oriented operation, whereas the detection training interface exposes the geometric and photometric transformations used here [21], [22]. Copy-paste [20] is discussed in Section II as augmentation background only; its original evaluation concerned instance segmentation and is not evidence for detection on this dataset.

Applied to the 70 selected images, the procedure produced 140 augmented images. Four source boxes required corner clipping to lie within the normalised image bounds, no source box was degenerate, no transformation failed, and 8 boxes were discarded across the generated variants under the visibility and area criteria. As required before training, a random sample of augmented images was rendered with their transformed boxes overlaid and inspected visually to confirm that box geometry tracked the image transformations; this check was performed and passed.

D. Combined Configuration and Dataset Composition

In the combined configuration the two components are composed as follows. Each selected image retains its original copy in the training set, and its three additional slots are filled with augmented variants first and with an exact duplicate once the two variants are exhausted. This yields 140 augmented images and 70 exact duplicates, so no selected image is replicated more than once without transformation. Fig. 2 summarises the pipeline from audit through grouping, manifest construction, targeted augmentation, training and evaluation.

E. Disclosure of the Dataset-Size Confound

Because oversampling and offline augmentation both increase the number of training images, the two configurations do not see the same number of images per epoch at a fixed epoch budget, and therefore do not take the same number of optimisation steps. The baseline configuration trains on 5,967 images per epoch and the class-aware configuration on

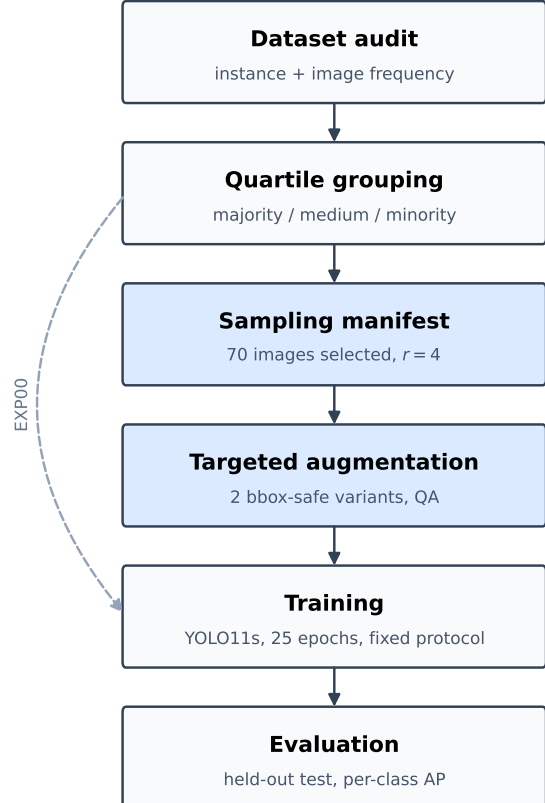


Fig. 2. Pipeline of the proposed method. The baseline configuration bypasses the sampling and augmentation stages and trains directly on the unmodified manifest.

6,177, an expansion factor of 1.035. Over the 25-epoch budget this corresponds to approximately 149,175 and 154,425 image presentations respectively. The step budget was not matched between configurations; the difference is disclosed here rather than corrected for, and its magnitude is reported so that it can be weighed against the observed differences in Section VII.

V. EXPERIMENTAL SETUP

A. Hardware and Software

All experiments were run on a single NVIDIA Tesla T4 with 15.6 GB of memory in a hosted notebook environment. The software stack comprised Python 3.13.15, PyTorch 2.11.0 with CUDA 12.8, cuDNN 9.19.0, Ultralytics 8.4.137, Albumentations 2.0.8 and NumPy 2.1.3. Automatic mixed precision was enabled and dataset caching was disabled, the latter because available system memory could not hold the dataset.

B. Training Protocol

Table II lists the training configuration, which is identical for both experiments apart from the training manifest. The optimiser was specified explicitly as SGD rather than left to automatic selection: the automatic setting derives the optimiser and learning rate from the size of the training set, which differs

TABLE II
TRAINING AND EVALUATION CONFIGURATION

Parameter	Value
Model	YOLO11s, COCO-pretrained [10]
Parameters	9,432,823
Input size	512 × 512
Epochs	25
Batch size	16
Optimiser	SGD, $lr_0 = 0.01$, momentum 0.937
Weight decay	0.0005
Warm-up epochs	3
Early stopping	Disabled
Random seed	42 (deterministic)
Mixed precision	Enabled
Dataset caching	Disabled
GPU	NVIDIA Tesla T4, 15.6 GB
Python / PyTorch / CUDA	3.13.15 / 2.11.0 / 12.8
Ultralytics / Albumentations	8.4.137 / 2.0.8
Train images per epoch	5,967 (EXP00); 6,177 (EXP03)
Training time	67.3 min (EXP00); 69.7 min (EXP03)
Random seeds	1

between configurations, and would therefore have introduced an uncontrolled difference. Early stopping was disabled so that both configurations completed the full 25-epoch budget. The random seed was fixed at 42 and deterministic execution was requested.

The input resolution was set to 512 rather than 640. A timing probe measured a single epoch on the larger of the two training manifests at 640 and found it exceeded the per-epoch budget available within the session limits of the environment; the reduced resolution was applied identically to both configurations. The training-time augmentation defaults of the framework — including mosaic, horizontal flipping, HSV jitter, scaling, translation and random erasing — were left at their library defaults and are identical across configurations. The targeted augmentation of Section IV is therefore additional to, not a replacement for, the framework’s own training-time augmentation.

One property of the initialisation warrants disclosure. The framework transfers classification head rows from the pre-trained checkpoint to target classes whose names match the pretraining vocabulary; the training log records that 6 of the 13 classes received such a transfer, with the remaining 7 initialised from scratch. This behaviour is deterministic, depends only on the class names, and is identical across both configurations, so the comparison remains controlled. It is nonetheless not uniform across the class groups defined in Section III, and Section VII considers its bearing on the per-class results.

C. Experimental Configurations

Two configurations were trained. EXP00 is the baseline: the unmodified training manifest with no oversampling and no targeted augmentation. EXP03 applies both components. The four-way ablation that would additionally isolate oversampling and augmentation separately was not run, owing to the compute budget available. Consequently no claim is made in this work as to which of the two components is responsible for any

TABLE III
OVERALL DETECTION PERFORMANCE ON THE BADODD TEST SPLIT

Config.	Samp.	Aug.	P	R	mAP ₅₀	mAP _{50:95}
EXP00 (baseline)	—	—	75.41	54.88	60.78	38.58
EXP03 (class-aware)	✓	✓	73.30	53.82	62.86	40.24
Δ			−2.10	−1.06	+2.08	+1.67

observed difference; the comparison is between the baseline and the combined strategy only, and the study is reported as preliminary in this respect.

D. Evaluation Protocol

The test split was held out entirely during training and configuration selection. Model selection within each run used validation mAP@0.50:0.95 to choose the retained checkpoint, and the test split was evaluated once, after both runs had completed. Evaluation used a confidence threshold of 0.001, a non-maximum-suppression IoU threshold of 0.7, batch size 16 and input size 512, identical for both configurations. Performance is reported as Precision, Recall, mAP@0.50 and mAP@0.50:0.95 with per-class AP and support counts, following standard detection metric definitions [23]. Latency was measured at batch size 1 over 200 test images on the same device for both configurations, under the consistent measurement conditions recommended for such comparisons [24]. A single random seed was used; no multi-seed replication was performed and no statistical significance testing is reported.

VI. RESULTS

This section reports the measured outcomes of the two configurations described in Section V. All values were obtained on the held-out test split of 2,033 images, which was evaluated once, after training of both configurations had completed. Detection performance is reported as Precision, Recall, mean Average Precision at IoU 0.50 (mAP@0.50), and mean Average Precision averaged over IoU thresholds from 0.50 to 0.95 (mAP@0.50:0.95), following standard detection metric definitions [23]. Precision and Recall are reported at the operating point selected by the evaluation procedure, whereas the mAP values integrate over confidence thresholds; the two families of metrics are therefore not directly comparable and are reported side by side.

A. Overall Detection Performance

Table III reports aggregate performance for the baseline configuration (EXP00) and the class-aware configuration (EXP03). The class-aware configuration recorded a higher mAP@0.50 (62.86 against 60.78, a difference of 2.08 percentage points) and a higher mAP@0.50:0.95 (40.24 against 38.58, a difference of 1.67 percentage points). Over the same comparison, Precision decreased by 2.10 percentage points and Recall decreased by 1.06 percentage points. The direction of change is therefore not uniform across the four aggregate metrics.

B. Per-Class Performance

Table IV reports per-class AP at both IoU settings alongside the training and test instance counts for each class. Of the thirteen classes, eight recorded a higher AP@0.50 under the class-aware configuration and five recorded a lower value. At AP@0.50:0.95, eight classes recorded a higher value and five a lower value, though the two sets are not identical: cart-vehicle recorded a higher AP@0.50 (+1.20) together with a lower AP@0.50:0.95 (−1.19), and person recorded a higher AP@0.50 (+0.16) together with a lower AP@0.50:0.95 (−0.11).

The largest recorded increases at AP@0.50 were construction-vehicle (+11.89), wheelchair (+6.80), and priority-vehicle (+5.56). The largest recorded decreases were truck (−0.65), car (−0.41), and three-wheeler (−0.27). At AP@0.50:0.95 the largest increase was wheelchair (+10.96) and the largest decrease was cart-vehicle (−1.19). Among the seven classes with more than 600 training instances, all changes at AP@0.50 fell within the interval [−0.65, +0.66] percentage points.

C. Group-Level Performance

Table V reports mean AP by the class groups defined in Section III. No class in this dataset copy met the extreme-support criteria, and every class had non-zero test support, so all thirteen classes are evaluable and the extreme group is empty. The majority group recorded a change of +0.03 percentage points at AP@0.50, the medium group +0.93, and the minority group +6.83. At AP@0.50:0.95 the corresponding changes were +0.17, +0.15, and +6.69.

D. Low-Support Class Observations

The three minority classes carry test supports of 11 (construction-vehicle), 16 (train), and 28 (wheelchair) instances. Because AP is computed over these counts, single detection outcomes translate into large changes in the reported value, and the per-class figures for this group carry substantially wider uncertainty than those of the remaining ten classes. No significance testing was performed and none of the differences reported for this group should be read as established effects.

Two specific observations are recorded. First, under the baseline configuration construction-vehicle recorded a Precision of 1.000 with a Recall of 0.000 and an AP@0.50 of 0.17; this combination indicates that essentially no detections of this class were retained at the reported operating point, and the Precision value for this cell should not be read as a precision-recall trade-off. Second, the train class recorded an AP@0.50 of 88.35 under the baseline configuration despite having only 21 training instances, which is higher than the value recorded for any of the three majority classes.

E. Training Behaviour

Fig. 3 shows the validation mAP trajectories for both configurations over the 25-epoch budget. Both configurations completed the full budget; early stopping was disabled so that the two runs received identical epoch counts.

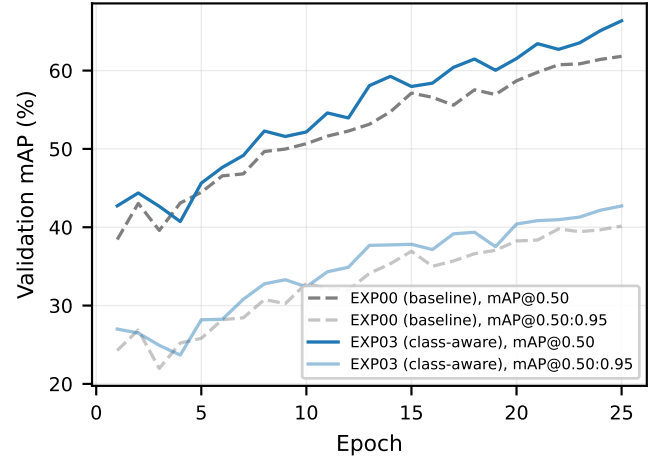


Fig. 3. Validation mAP@0.50 and mAP@0.50:0.95 across the 25-epoch budget for both configurations.

F. Qualitative Examples

Fig. 4 shows paired detection outputs from the two configurations on test images containing minority-group objects, rendered at a confidence threshold of 0.25.

G. Efficiency

Both configurations use the identical YOLO11s architecture and differ only in training data, so the parameter count (9,432,823) and checkpoint size (19.15 MB) are identical. Measured on the same Tesla T4 under identical conditions [24], single-image inference latency was 22.01 ms (45.4 FPS) for the baseline configuration and 21.48 ms (46.6 FPS) for the class-aware configuration, timed over 200 test images at batch size 1 and an input size of 512. As the two configurations produce architecturally identical models, this 0.53 ms difference reflects measurement variation rather than a property of either configuration, and a separate efficiency table is therefore omitted. Wall-clock training time was 67.3 minutes for the baseline configuration and 69.7 minutes for the class-aware configuration.

VII. DISCUSSION

A. Did the Class-Aware Strategy Improve Minority-Category Detection?

Observation. The minority group, comprising construction-vehicle, train and wheelchair, recorded a mean AP@0.50 of 47.74 under the baseline and 54.56 under the class-aware configuration, a difference of 6.83 points, with a corresponding difference of 6.69 points at AP@0.50:0.95. The majority group changed by 0.03 points and the medium group by 0.93 points at AP@0.50.

Interpretation. The difference is concentrated in the group to which the interventions were applied, and the largest single contribution comes from construction-vehicle, whose AP@0.50 moved from 0.17 to 12.06. The magnitude of that change should not be mistaken for competence: a value of

TABLE IV
PER-CLASS AVERAGE PRECISION WITH TRAINING AND TEST SUPPORT

Class	Train	Test	AP ₅₀ base	AP ₅₀ prop.	Δ AP ₅₀	AP _{50:95} base	AP _{50:95} prop.	Δ AP _{50:95}
person	18,268	6,325	68.93	69.09	+0.16	39.53	39.43	-0.11
auto-rickshaw	10,642	3,517	85.81	86.02	+0.20	61.86	62.45	+0.59
three-wheeler	5,732	1,984	82.39	82.12	-0.27	57.70	57.72	+0.02
car	3,786	1,236	78.85	78.43	-0.41	54.00	53.66	-0.34
motorbike	3,771	1,266	71.60	71.88	+0.28	37.16	38.20	+1.04
truck	2,304	782	70.53	69.88	-0.65	46.00	45.48	-0.52
bus	1,885	620	73.19	73.86	+0.66	49.61	50.09	+0.48
bicycle	678	256	44.87	44.74	-0.13	20.90	20.86	-0.04
priority-vehicle	230	66	51.62	57.18	+5.56	38.22	39.86	+1.64
cart-vehicle	144	46	19.12	20.32	+1.20	11.57	10.38	-1.19
wheelchair [†]	29	28	54.68	61.48	+6.80	36.72	47.68	+10.96
construction-vehicle [†]	24	11	0.17	12.06	+11.89	0.12	6.07	+5.95
train [†]	21	16	88.35	90.15	+1.79	48.15	51.30	+3.16

[†] Minority group; test support below 30 instances (Section VI-D). All values in percent; Δ in percentage points.

TABLE V
GROUP-LEVEL MEAN AVERAGE PRECISION

Group	n	mAP ₅₀ b.	mAP ₅₀ p.	Δ	mAP _{50:95} b.	mAP _{50:95} p.	Δ
Majority	3	79.05	79.08	+0.03	53.03	53.20	+0.17
Medium	7	58.54	59.47	+0.93	36.78	36.93	+0.15
Minority [†]	3	47.74	54.56	+6.83	28.33	35.02	+6.69
Extreme	0	—	—	—	—	—	—

[†] Combined test support of 55 instances.

12.06 represents a detector that fails on this category most of the time. The baseline result of 0.17, accompanied by a recall of zero, indicates that essentially no construction-vehicle detections survived the reporting threshold. The difference between the two configurations for this category is therefore a difference between complete failure and near-complete failure, not between failure and adequacy.

The group mean is also sensitive to composition. The train category, with a baseline AP@0.50 of 88.35, contributes disproportionately to the group average. Excluding it, the mean AP@0.50 of the remaining two minority categories moves from 27.43 to 36.77, a difference of 9.34 points. Both framings are reported because neither alone conveys the state of the group.

B. Was There a Cost to Frequent Categories?

Observation. At group level the majority categories changed by 0.03 points at AP@0.50 and 0.17 points at AP@0.50:0.95. At category level, changes among the seven categories with more than 600 training instances all fell within the interval from -0.65 to +0.66 points at AP@0.50.

Interpretation. No meaningful degradation of frequent-category performance was observed. This is consistent with the design of the intervention, which modified 70 of 5,967 training images and left the remainder untouched. The concern that oversampling rare categories degrades common ones did not materialise at this intervention strength; it is not established that it would not at a stronger one.

C. Why Did Precision and Recall Fall While mAP Rose?

Observation. Aggregate precision fell by 2.10 points and recall by 1.06 points, while both mAP measures rose. Examined

per category, precision rose for eleven of thirteen categories and recall fell for ten.

Interpretation. The aggregate precision decline is an artefact of a single category. The baseline recorded a precision of 1.000 for construction-vehicle at a recall of zero, which is the degenerate value produced when almost no detections are retained. Excluding that category, mean precision rises from 73.36 to 77.54, a difference of 4.19 points, while mean recall still falls from 59.45 to 57.55, a difference of 1.91 points. The class-aware configuration therefore operates more conservatively at the selected threshold, making fewer but more frequently correct predictions, while its higher mAP indicates better ranking of detections across the full range of confidence values. Reporting the aggregate precision figure alone would misrepresent this, which is why both are given in Section VI.

D. Can the Difference Be Attributed to the Intervention?

Observation. The priority-vehicle category, which was not targeted by either intervention, recorded a change of +5.56 points at AP@0.50 — comparable in magnitude to wheelchair at +6.80 and larger than train at +1.79. The cart-vehicle category, also untargeted, recorded +1.20 at AP@0.50 together with -1.19 at AP@0.50:0.95.

Interpretation. This is the most important qualification in the study. If categories that received no intervention can move by amounts comparable to those that did, then run-to-run variation is of the same order as the effect being measured, and the per-category differences cannot be separated from it on the evidence available. A single training seed was used and no significance testing was performed, so no statement about the reliability of individual category differences is warranted.

Hypothesis. Two observations argue against attributing the group-level pattern entirely to variation, though neither settles the matter. First, the majority group changed by 0.03 points, which is what would be expected if the intervention acted locally rather than through a general effect of additional optimisation steps; had the 3.5 percent increase in image presentations been the operative cause, a broader pattern of change would be anticipated. Second, the direction of change



Fig. 4. Qualitative comparison on test images containing minority-group objects. Left column: baseline configuration. Right column: class-aware configuration. Detections rendered at a confidence threshold of 0.25.

in the minority group is consistent at both IoU thresholds and across all three of its categories. These are indications, not evidence, and the appropriate test is multi-seed replication.

E. Verdict on the Hypotheses

The results are consistent with H1, which holds that increased exposure to images containing under-represented categories, combined with targeted augmentation of those images, improves minority-category AP relative to a reproducible baseline. They do not establish it. The null hypothesis H0, which

holds that the strategy does not improve minority performance beyond ordinary training variation, cannot be rejected on this evidence, because the study provides no estimate of that variation. The observed pattern is suggestive and the direction is consistent, but a single seed and a single dataset copy do not support a stronger claim.

A further limitation of attribution is structural. Because the compute budget permitted only the baseline and the combined configuration, the contributions of oversampling

and of targeted augmentation cannot be separated. Any of three explanations remains available: that oversampling alone accounts for the difference, that augmentation alone does, or that the two act jointly.

F. Interpreting the Rare Categories

Observation. The train category recorded an AP@0.50 of 88.35 under the baseline from 21 training instances, exceeding every majority category, with a recall of 0.875 that was identical under both configurations.

Hypothesis. Three factors plausibly contribute. Trains are large, visually distinctive and unlikely to be confused with other categories in the label set. The category is one of those whose classification head weights were initialised by name-match transfer from the pretrained checkpoint, as described in Section V, giving it an initialisation advantage not shared by the other two minority categories. And with 16 test instances concentrated in 16 images, the measurement itself is coarse. This last point applies to the whole minority group: at supports of 11, 16 and 28 instances, individual detection outcomes shift AP by several points, and the interventions studied here cannot create information that a 21-instance category does not contain.

VIII. LIMITATIONS

The scope of the claims made in this paper is limited in several specific ways, stated here rather than left to inference.

Statistical. Each configuration was trained once, with a single random seed. No multi-seed replication was performed and no significance testing is reported. As Section VII notes, untargeted categories moved by amounts comparable to targeted ones, which indicates that run-to-run variation may be of the same order as the measured effect. No difference reported in this paper should be treated as an established effect.

Experimental design. Only the baseline and the combined configuration were trained. The two intermediate configurations that would isolate oversampling from targeted augmentation were not run, so the study cannot attribute its findings to either component. It is reported as a preliminary two-arm comparison, not as an ablation.

Training budget. A 25-epoch budget was used and the input resolution was reduced from 640 to 512 after a timing probe showed the larger resolution exceeded the available per-epoch budget. Neither configuration was trained to convergence, and the ordering of the two configurations under a longer budget or a larger input size is unknown.

Dataset-size confound. Oversampling and offline augmentation increase the number of training images, so the class-aware configuration trained on 6,177 images per epoch against the baseline’s 5,967, an expansion factor of 1.035, and consequently took approximately 3.5 percent more optimisation steps. The step budget was not matched. This difference is small but is not zero, and it was disclosed rather than corrected.

Support. The three minority categories carry test supports of 11, 16 and 28 instances. AP computed over such counts is

coarse, and category-level differences within this group carry wide uncertainty.

Initialisation. Classification head weights were transferred by name match from the pretrained checkpoint for 6 of 13 categories. This is identical across configurations and does not affect the comparison, but it is not uniform across the category groups and confers an initialisation advantage on some categories that others do not receive.

Dataset provenance. The publicly distributed release used here differs from the statistics reported in the original description, containing 207 more images and 1,093 more annotated objects, with two categories differing by an order of magnitude in training support. The class-name mapping was not distributed with the release and was recovered and verified as described in Section III. Results reported here describe this release, and comparison with figures reported elsewhere for BadODD should account for this.

Generality. A single detector at a single scale was evaluated on a single dataset. No claim is made that these findings extend to other detectors, other scales, or other datasets. No latency or real-time deployment claim is made beyond the measurements reported in Section VI, which were taken on one device under one set of conditions.

IX. CONCLUSION

This paper asked whether class-aware image sampling and targeted data augmentation can improve detection of under-represented categories in BadODD while maintaining overall performance. Two configurations of a fixed YOLO11s detector were trained under an identical protocol and evaluated once on a held-out test split. The combined strategy raised mean average precision at IoU 0.50 from 60.78 to 62.86 and at IoU 0.50:0.95 from 38.58 to 40.24, with the difference concentrated in the targeted category group, which gained 6.83 points at IoU 0.50 against 0.03 points for the most frequent categories. Aggregate precision and recall declined, though the precision decline was traced to a single degenerate baseline measurement.

These results are consistent with the hypothesis that class-aware training improves minority-category detection without a cost to frequent categories, but they do not establish it: untargeted categories moved by comparable amounts, and a single training seed provides no estimate of run-to-run variation. Three directions follow directly. Multi-seed replication would establish whether the observed differences exceed training variation. Separate runs would isolate the contributions of sampling and augmentation. And comparison against loss-reweighting methods would situate this data-centric approach among its alternatives.

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